BT-202 (GS)

B.Tech., I & II Semester Examination, December 2025

Grading System (GS)

MATHEMATICS-II

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8)

1. (a) Solve: $(1 + y^2)\,dx = (\tan^{-1} y - x)\,dy$. **7**

(b) Solve the simultaneous differential equations: $d^2x/dt^2 - 4\,dx/dt + 4x = y$ and $d^2y/dt^2 + 4\,dy/dt + 4y = 25x + 16e^t$. **7**

2. (a) Find the series solution of xy'' + y' + xy = 0 about x = 0 by the Frobenius method. **7**

(b) Apply the method of variation of parameters to solve: $d^2y/dx^2 + y = \tan x$. **7**

3. (a) Solve: (D² − 2DD' + D'²) Z = e^(x + 2y) + x². **7**

(b) Find a complete integral of px + qy = pq. **7**

4. (a) Show that $u = e^x (x \cos y - y \sin y)$ is a harmonic function. Find the conjugate function v. **7**

(b) Evaluate ∮ e^z / [(z − 1)(z − 4)] dz where C is the circle |z| = 2, by using Cauchy's Integral Formula. **7**

5. (a) Find the directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at the point P(2, 1, 3) in the direction of the vector $\vec{a} = \hat{i} - 2\hat{k}$. **7**

(b) Evaluate ∮ F·dr, where $F = x^2\,\hat{i} + y^3\,\hat{j}$ and the curve C is the arc of the parabola $y = x^2$ in the xy-plane from (0, 0) to (1, 1). **7**

6. (a) Evaluate the integral using residue theorem: ∮ (4 − 3z) / [(z − 1)(z − 2)] dz where C is the circle |z| = 3/2. **7**

(b) Evaluate the integral ∫₀^(2π) dθ / (5 − 3 cos θ). **7**

7. (a) Verify Stokes' theorem for the vector field $F = (x^2 - y^2)\,\hat{i} + 2xy\,\hat{j}$, integrated around the rectangle z = 0 bounded by the lines x = 0, y = 0, x = a and y = b. **7**

(b) Define Gradient, Divergence and Curl. **7**

8. (a) Solve: (D² − D'² − 3D + 3D') Z = e^(x + 2y) + xy. **7**

(b) By reducing to homogeneous form, solve: $(1 + x)^2\,d^2y/dx^2 + (1 + x)\,dy/dx + y = 4\cos\{\log(1 + x)\}$. **7**

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